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SYLLABUS

PN Junction Diode and Diode Circuits: Semiconductor Materials, Construction of PN Junction, Formation of Depletion Layer and Barrier Voltage in PN Junction, Current-Voltage Characteristics of a PN Junction Diode, Equivalent Circuit of Diode, Resistance and Capacitance of PN Junction, Diode Circuit and Load Line, Zener Diode Operation and Application, Half-Wave and Full-Wave Rectifier, Voltage Multiplier, Clipper and Clamper.

Bipolar Junction Transistors (BJT): Construction and Operation of BJT, Amplifying Action, Characteristics of BJT in Common Base (CB), Common Emitter (CE) and Common Collector (CC) Configurations, α and β , Q-Point and Load Line, Different Biasing Circuits, Stability Factor, BJT as a Switch, re and Hybrid Equivalent Circuit of BJT.

Single Stage BJT Amplifier Circuits: Operation of Single-Stage Amplifier, Voltage and Current Gain, Input and Output Impedance of CB, CE, CC Configurations using h-Parameter, Effect of Unbypassed R_E, R_S.

Field Effect Transistor (FET): JFET Structure, Operation and Characteristics, hParameters for JFET, MOSFET Construction, Operation and Characteristics, Biasing Circuits of JFET and MOSFET, Single-stage JFET Amplifier, MOSFET as a Switch, CMOS Inverter.

Multistage and Differential Amplifiers: RC Coupled Two Stage Amplifiers, Direct-coupled Amplifier, Darlington Pair, Multistage Frequency Effects.

Feedback Amplifiers: Principle of Feedback Amplifier, Positive and Negative Feedback, Advantages of Negative Feedback, Gain Stability, Decreased Distortion, Increased Bandwidth, Forms of Negative Feedback, Practical Feedback Circuits.

REFERENCE BOOKS

1. Electronic Devices and Circuits, 5^{th} Edition, David Bell, Oxford University Press.
2. Electronic Devices and Circuit Theory, Paperback Edition, R. Boylestad and L.Nashelsky, Pearson.
3. Electronic Devices and Circuits, Allen Mottershead, Goodyer Pub. Co.
4. Electronic Principles, Paperback Edition, Albert Malvino and David Bates, McGraw Hill.
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CONTENTS

CHAPTER 1

PN JUNCTION DIODE AND DIODE CIRCUITS

1.1 Semiconductor Materials

Conductor: The term conductor is used for any material that will support generous flow of charge when the voltage is applied across its terminals. Ex:Cu,Al etc.

Insulator: An insulator is a material that offers very low conductivity when voltage is applied. Ex : Mica,Rubber,Wood etc.

Semiconductor: It is a material that has conductivity more than insulators but less than the conductor. Ex: Si, Ge

Resistivity and Conductivity

$$\rho = \frac{1}{\sigma} = \frac{RA}{L}$$

Where,

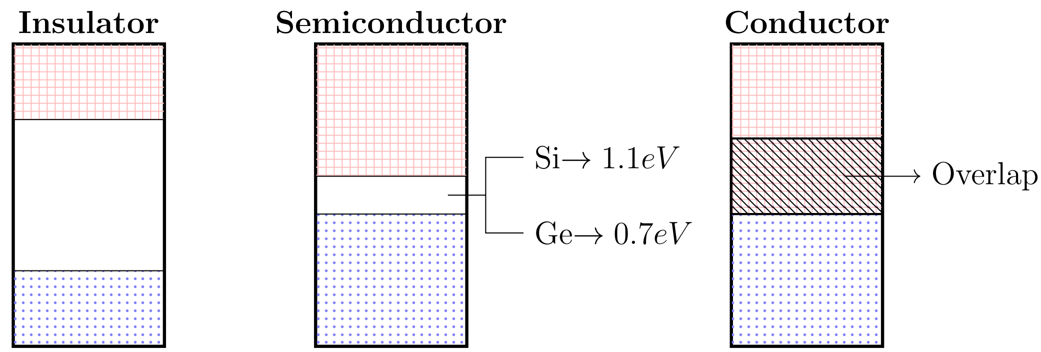
ρ = Resistivity [Its a property of material]
σ = Conductivity
A = Area of the cross-section
R = Resistance
L = Length

According to Ohm's Law: $R \uparrow I \downarrow$ and $R \downarrow I \uparrow$

So,

$$\begin{aligned} \rho_c < \rho_s < \rho_i \qquad & \rho \longrightarrow \text{Resistivity} \\ \sigma_c > \sigma_s > \sigma_i \qquad & \sigma \longrightarrow \text{Conductivity} \\ I_c > I_s > I_i \qquad & I \longrightarrow \text{Current} \end{aligned}$$

Energy Band Diagram



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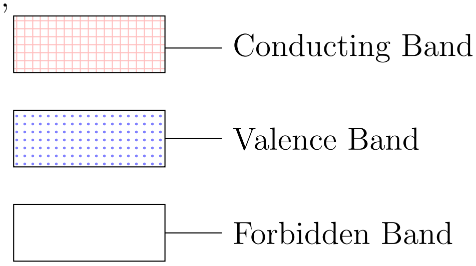


Crystal Structure of Silicon

- Silicon crystallize in a diamond cubic lattice. Each Si has 4 valence electrons. To achieve stability, each atom shares 1 valence electron with each of its four neighbours.

Intrinsic Semiconductor

- They are the pure semiconductors. No other atoms included.
- Free electrons are only due to natural causes.
- At $T = 0K$ temperature, all bonds are intact, with no free electrons for conduction.
- At $T \neq 0K$,some covalent bonds break where a broken covalent bond frees an electron, allowing it to move and conduct current when an electric field is applied.

Q 1.0: Why do we bother with the concept of the hole?

We bother with holes because they simplify the description of current flow in semiconductors. Instead of tracking billions of electrons in the valence band, we treat the absence of one electron as a positive charge carrier. This makes equations (like drift current, diffusion current, pn-junction equations, etc.) symmetric and elegant.Device physics depends on holes (like PN junctions (diodes), BJTs) many modern devices rely on the interaction of electrons and holes.

$$\text{Current in a diode} = \text{electrons injected one way} + \text{holes injected the other way}$$

Even though holes are "not real," they behave as if they are real positive particles. They have measurable mobility, effective mass, and contribute to measurable current density. Hall-effect experiments even let us measure whether the dominant carriers are electrons

or holes (n-type vs. p-type).
So holes are not imaginary – they are a useful quasi-particle.

Q 1.1: Why are holes slower than electron?

Holes are slower than electrons because they have a larger effective mass and experience more scattering in the valence band.Electrons occupy the conduction band, which has a simpler, parabolic energy-momentum relation.The E−k relation (energy vs. crystal momentum):

$$\frac{1}{m^*} = \frac{1}{\hbar^2}\frac{dE^2}{dk^2}$$

- If the band is very curved (steep parabola) $\rightarrow \dfrac{dE^2}{dk^2}$ is large $\rightarrow$ small effective mass.
- If the band is flatter (gentle parabola) $\rightarrow \dfrac{dE^2}{dk^2}$ is small $\rightarrow$ large effective mass.

For e^- effective mass is small $\rightarrow$ they accelerate easily.
On the other hand holes are in the valence band, which is more complex (split into heavy-hole and light-hole bands). The average effective mass of holes is larger. Bigger effective mass $\rightarrow$ lower mobility.Again holes interact more strongly with the crystal lattice because of their complex band structure.
This increases collisions (phonon and impurity scattering), reducing their mobility further.

Extrinsic Semiconductor

- Impurity atoms are added.
- Two types of impurity
  - Pentavalent Impurity: Added atoms like Antimony, P, As
  - Trivalent Impurity: Added atoms like B, Al
- Added 1 part in 10 millions.
- **Doping:** Process of adding certain impurity atoms to the pure semiconductors is called doping.
- There are **2** types of extrinsic semiconductor.
  - **n-type:** we added pentavalent impurity.
  - **p-type;** we added trivalent impurity.

Directions:

$$\xrightarrow{\quad e^- \quad}$$

$$\xleftarrow{\quad I\text{(Conventional Current)} \quad}$$

$$\xrightarrow{\quad I'\text{(Natural Current)} \quad}$$

$$\xleftarrow{\quad \text{hole} \quad}$$

Properties Name	**Si(Silicon)**	**Ge (Germanium)**
Energy Gap (eV)	1.1	0.7
Electron Mobility ($m^2/V.s$)	0.135	0.39
Hole Mobility ($m^2/V.s$)	0.048	0.19
Intrinsic Carrier Density (n_i)	1.5×10^{16}	2.4×10^{19}
Intrinsic Resistivity	2300	0.46
Density (gm/m^3)	2.33×10^6	3.32×10^6

1.2 Mass Action Law

Statement: Under thermal equilibrium, the product of the free electrons concentration and the free hole concentration is equal to a constant to the square of intrinsic carrier concentration.
In Intrinsic Semiconductor,

no of e^- = no of holes
$n = p = n_i$

$$n.p = {n_i}^2 \quad \left|\begin{array}{l} n = \text{Free electron concentration} \\ p = \text{Free hole concentration} \\ n_i = \text{Intrinsic Carrier Concentration} \end{array}\right.$$

<u>Prove :</u>
Because of thermal energy ,the e^- will break the covalent bond and participate in conduction leaving the hole behind and generate charge carriers. Under the thermal equilibrium.

Rate of Generation = Rate of Recombination

$$G = R$$

$$R \propto n$$
$$R \propto p$$

$$R = rnp \qquad [\text{r} \rightarrow \text{Recombination constant}]$$

In intrinsic semiconductor, $G = rn_i^2$. Dopping in intrinsic semiconductor is not affected by the generation of holes or electrons.

1.3 Charge Density in a Semiconductor

- Electrical neutrality is satisfied by semiconductor also.

$$\text{total} \oplus \text{charge} = \text{total} \ominus \text{charge}$$

$$N_D + p = N_A + n$$

N_A = Concentration of Acceptor ion

N_D = Concentration of Donor ion

p = Hole Concentration

$n = e^-$Concentration

Case 1: Consider n-type material

Here,

$$N_A = 0$$

So,

$$N_D + p = n$$

$$\Rightarrow N_D = n - p$$

$$n >> p$$

$$N_D = n$$

$$n_n \simeq N_D$$

$$n.p = {n_i}^2$$

$$\Rightarrow n_n.p_n = {n_i}^2$$

$$\Rightarrow N_D.p_n = {n_i}^2$$

$$\Rightarrow p_n = \frac{{n_i}^2}{N_D}$$

Case 2: Consider p-type material

Here,

$$N_D = 0$$

So,

$$N_A + n = p$$

$$\Rightarrow N_A = p - n$$

$$p >> n$$

$$p = N_A$$

$$p_p \simeq N_A$$

$$n.p = {n_i}^2$$

$$\Rightarrow n_p.p_p = {n_i}^2$$

$$\Rightarrow n_p.N_A = {n_i}^2$$

$$\Rightarrow n_p = \frac{{n_i}^2}{N_A}$$

1.4 Conductivity of Semiconductor

- Metals are unipolar. But semiconductors are bipolar.
- When Electric Field E is applied, e^- and holes are drifted in opposite direction bu the current due to them will be in the same direction.

$$v_d \propto E$$
$$\Rightarrow v_d = \mu E$$
$$i = neAv_d$$
$$\sigma = ne\mu = (n\mu_n + p\mu_p)e$$
$$J = \frac{i}{A} = nev_d = ne\mu E = \sigma E = (n\mu_n + p\mu_p)eE$$

μ = Mobility
i = Current
n = no. of charge carrier
A = Area of cross-section
J = Current Density
σ = Conductivity
E = Electric Field
v_d = Drift Velocity
e = Charge of a e^-
$e.m$ = Effective Mass

$$\mu \propto \frac{1}{e.m}$$

$e.m.\ of\ hole >> e.m.\ of\ e^-$ $\quad \mu_p << \mu_n$

Example 1.1: The mobilities of free electron and holes in a pure germanium are 0.38 and 0.18 $m^2/V.s$. Find the value of intrinsic conductivity. Assume $n_i = 2.5 \times 10^{19}/m^3$ at room temperature.

Given,

Mobility of free electron, $\mu_n = 0.38/m^3$
Mobility of holes, $\mu_p = 0.18/m^3$

$\because$This is a pure germanium semiconductor.

$$\therefore n = p = n_i = 2.5 \times 10^{19}/m^3$$

We know that,

$$\text{Conductivity, } \sigma = (n\mu_n + 0\mu_p)e$$
$$= (0.38 + 0.18) \times 2.5 \times 10^{19} \times 1.6 \times 10^{-19} (\Omega - m)^{-1}$$
$$= 2.24\ (\Omega - m)^{-1}{}_{(ans)}$$

Question 1.1: In a germanium sample, a donor type impurity is added to the extent of 1 atom per 10^8 germanium atoms. Show that resistivity of the germanium sample drops to 3.7 ohm-cm
Given,

$$\mu_n = 3800cm^2/V.s \qquad n_i = 2.5 \times 10^{13}/cm^3$$

$$\mu_p = 1800cm^2/V.s \qquad N_{ge} = 4.41 \times 10^{22}/cm^3$$

Given,

Mobility of free electron: $\mu_n = 3800\,\text{cm}^2/\text{V·s}$
Mobility of holes: $\mu_p = 1800\,\text{cm}^2/\text{V·s}$
Concentration of intrinsic charge carrier: $n_i = 2.5 \times 10^{13}\,\text{cm}^{-3}$
No of Germanium e^- per atom: $N_{Ge} = 4.41 \times 10^{22}\,\text{cm}^{-3}$

For doping semiconductor,

$$N_D = \frac{N_{Ge}}{\text{Extent electron per atom}} = \frac{4.41 \times 10^{22}}{10^8} = 4.41 \times 10^{14}\,\text{cm}^{-3}$$

$n = N_D = 4.41 \times 10^{14}\,\text{cm}^{-3}$

$$p = \frac{n_i^2}{n} = \frac{(2.5 \times 10^{13})^2}{4.41 \times 10^{14}} = 1.51 \times 10^{12}\,\text{cm}^{-3}$$

We know,

$$\rho = \frac{1}{\sigma} = \frac{1}{(n\mu_n + p\mu_p)e}$$

$$= \frac{1}{(4.41 \times 10^{14} \times 3800 + 1.51 \times 10^{12} \times 1800) \times 1.6 \times 10^{-19}}$$

$$= 3.723\,\Omega \cdot \text{cm}$$

Question 1.2: Consider an intrinsic silicon bar of cross-section $5\,\mathrm{cm}^2$ and length $0.5\,\mathrm{cm}$ at room temperature 300K. An average field of $20\,\mathrm{V/cm}$ is applied across the ends of the silicon bar.

Calculate:

a) Electron and hole components of current density.
b) Total current in the bar.
c) Resistivity of the bar.

Given,

$\mu_n = 1400\,\mathrm{cm}^2/\mathrm{V{\cdot}s}$ $\quad n_i = 1.5 \times 10^{10}/\mathrm{cm}^3$

$\mu_p = 450\,\mathrm{cm}^2/\mathrm{V{\cdot}s}$

Given,

No of intrinsic charge carrier $n_i = 1.5 \times 10^{10}/\mathrm{cm}^3$
charge $q = 1.6 \times 10^{-19} C$
Electric field $E = 20\,V cm^{-1}$
Mobility of holes $\mu_p = 450\,\mathrm{cm}^2/\mathrm{V{\cdot}s}$
Mobility of electron $\mu_n = 1400\,\mathrm{cm}^2/\mathrm{V{\cdot}s}$
charge density of holes $J_p =?$
Charge density of electron $J_n =?$

(a) We know,

$$J_n = \mu_n nqE = 1400 \times 1.5 \times 10^{10} \times 1.6 \times 10^{-19} \times 20 = 6.72 \times 10^{-5}\ Acm^{-2}$$

$$J_p = \mu_p nqE = 450 \times 1.5 \times 10^{10} \times 1.6 \times 10^{-19} \times 20 = 2.16 \times 10^{-5}\ Acm^{-2}$$

(b)We know

$$\begin{aligned} J_{total} &= J_n + J_p \\ &= 6.72 \times 10^{-5} + 2.16 \times 10^{-5} Acm^{-2} \\ &= 8.8827 \times 10^{-5} Acm^{-2} \end{aligned}$$

Now,

$$I = JA = 8.8827 \times 10^{-5} \times 5\ A = 444.1\ \mu A$$

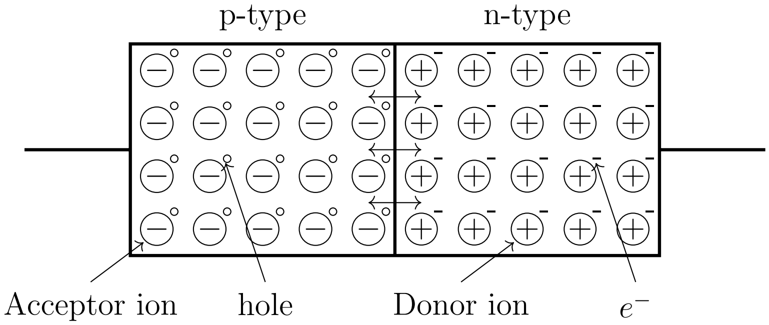
(c)We know

$$\rho = \frac{1}{\sigma} = \frac{1}{(n\mu_n + p\mu_p)e}$$

$$= \frac{1}{(1.5 \times 10^{10} \times 1400 + 1.5 \times 10^{10} \times 450) \times 1.6 \times 10^{-19}}$$

$$= 2.25 \times 10^{5} \Omega \cdot cm$$

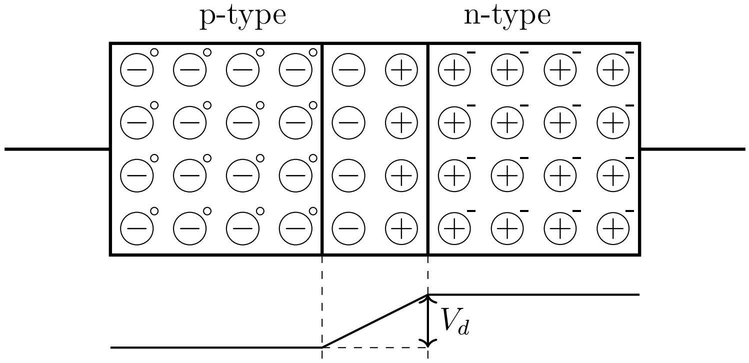
1.5 PN Junction Diode (No Applied Bias)



- Bias means application of external voltage across the two terminals.
- BJT means Bipolar Junction Transistor (3 terminals device, having 2 diodes)
- PN Junction constitutes a rectifier which permits the easy flow of charge in one direction but restrains the flow in the opposite direction.
- p-type
  - Majority Charge Carrier: hole
  - Minority Charge Carrier: e^-
- n-type
  - Majority Charge Carrier: e^-
  - Minority Charge Carrier: hole
- Diffusion current $\longrightarrow$ because of Majority Charge Carrier
- Immobile ions will surface out because of diffusion, it it depletion region.
- Depletion Region/Space Charge Region/ Depletion Layer : No mobile charge but only uncovered immobile ions.
- Width of depletion region is **fixed**.
- Drift current $\longrightarrow$ because of Minority Charge Carrier
- Diffusion current have the opposite direction of drift current.
- Under Steady State:

$$\text{Diffusion Current} = \text{Drift Current}$$
$$\text{Net current } = 0$$

1.6 Barrier Potential



Barrier Potential/ Built-in Potential,

$$V_b = \frac{kT}{e} \ln\left(\frac{N_A N_D}{{n_i}^2}\right)$$
$$= V_T \ln\left(\frac{N_A N_D}{{n_i}^2}\right)$$

Here,

k = Boltzman constant = $1.38066 \times 10^{-23} J/K$
T = Absolute Temperature
e = Charge of one $e^{-} = 1.6 \times 10^{-19} C$

At room temperature (300K/27°C):

$$V_T = \frac{kT}{e}$$
$$= \frac{1.38066 \times 10^{-23} \times 300}{1.6 \times 10^{-19}}$$
$$= 0.026V$$

• In ideal cases, we consider the barrier potential for Si is 0.7V and 0.3V is for Ge at room temperature.

Example 1.2: Consider a pn junction at room temperature, doped ar $N_A = 10^{16}/cm^3$ in the p-region and $N_D = 10^{17}/cm^3$ in the n-region. Intrinsic carrier density is $1.5/cm^3$ at room temperature. Calculate the built-in potential.

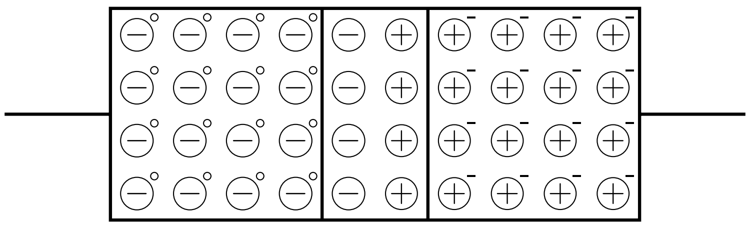
Given,

Acceptor ions Concentration, $N_A = 10^{16}/cm^3$
Donor ions Concentration, $N_D = 10^{17}/cm^3$
At room temperature, $V_T = 0.026V$
Intrinsic Charge density, $n_i = 1.5/cm^3$

We know that,

$$\begin{aligned}\text{Built-in Potential, } V_b &= V_T \ln\left(\frac{N_A N_D}{{n_i}^2}\right)\\ &= 0.026 \ \ln\left\{\frac{10^{16}\times 10^{17}}{(1.5\times 10^{10})^2}\right\}\\ &= 0.757V\end{aligned}$$

1.7 Width of Depletion Layer



Width of Depletion Region,

$$W_d = \sqrt{\frac{2\varepsilon}{e}\left(\frac{1}{N_A}+\frac{1}{N_D}\right)V_b}$$

$$\varepsilon = \varepsilon_r\varepsilon_0 \quad \left|\begin{aligned}&\varepsilon = \text{Electric Permittivity}\\ &\varepsilon_r = \text{Relative Permittivity}\\ &\varepsilon_0 = \text{Permittivity of Vacuum}\end{aligned}\right.$$

- For Si
  - $\varepsilon = 1.04\times 10^{-12}F/cm$
  - $0.1\mu m < W_d < 1\mu m$
- For Ge, $\varepsilon = 1.12448\times 10^{-12}F/cm$

Example 1.3: Consider a silicon pn-junction at room temperature doped at $N_A = 10^{16}/cm^3$ in the p-region and $N_D = 10^{17}/cm^3$ in the n-region. Intrinsic carrier density is $1.5/cm^3$ at room temperature. Calculate the width of depletion region.

Given,

Acceptor ions Concentration, $N_A = 10^{16}/cm^3$
Donor ions Concentration, $N_D = 10^{17}/cm^3$

At room temperature, $V_T = 0.026V$
Intrinsic Charge density, $n_i = 1.5/cm^3$
Permittivity of Si, $\varepsilon = 1.04 \times 10^{-12}$ F/cm

We know that,

$$\begin{aligned}
\text{Built-in Potential, } V_b &= V_T \ln\left(\frac{N_A N_D}{{n_i}^2}\right) \\
&= 0.026 \ln\left\{\frac{10^{16} \times 10^{17}}{(1.5 \times 10^{10})^2}\right\} \\
&= 0.757V
\end{aligned}$$

$$\begin{aligned}
\text{Width of depletion region,} W_d &= \sqrt{\frac{2\varepsilon}{e}\left(\frac{1}{N_A} + \frac{1}{N_D}\right).V_b} \\
&= \sqrt{\frac{2 \times 1.04 \times 10^{-12}}{1.6 \times 10^{-19}}\left(\frac{1}{10^{16}} + \frac{1}{10^{17}}\right) \times 0.757} \\
&= 3.290 \times 10^{-5}\text{cm} \\
&= 0.3290\mu\text{m}
\end{aligned}$$